Detect, Refine, Resolve: An Edge-Attention-Inspired Boundary Refiner for Solar Filament Instance Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. Our pipeline detects filament candidates with a YOLO11-seg instance detector, corrects their boundaries with a crop-level U-Net refiner, and resolves overlapping candidates into a disjoint instance set. Major design decisions were evaluated under leak-free 3-fold cross-validation with an observation-cluster bootstrap for uncertainty estimates, a protocol that let us reject five plausible improvements whose apparent single-fold gains did not survive independent-fold confirmation. One further change — augmenting the refiner's bottleneck with an EdgeAttNet-inspired [9] edge-guided self-attention mechanism (Section 2.3) — initially passed this protocol (3-fold pooled ΔPQ= +0.0028, 95% CI $[0.00091, 0.00476]$, n=707, all folds agreeing in sign) and moved the public leaderboard score from 0.37 to 0.39. A follow-up matched-parameter ablation across all three folds (two seeds each; Section 3.2) refines this picture rather than simply reversing it: bottleneck attention capacity, with or without the edge prior, tends to help over the plain baseline (a seed-consistent positive delta in two of three folds, roughly neutral in the third), but the edge prior itself shows no consistent additional benefit over plain attention, at or below it in two of three folds. We report this refinement, not the original single-seed claim, as our main finding: our acceptance criterion could not separate a general architectural effect (added capacity) from the specific mechanism (the edge prior) we set out to validate.

1 Introduction

This report describes our submitted solution to the Solar Filament Segmentation Challenge 2026 [3], organized by the Earth-Space AI Research (ESAIR) Lab and scored against the MAGFiLO dataset [2]. Our approach treats the task as instance segmentation: detect each filament as a separate object, refine its boundary, and reconcile overlapping candidates into a single non-overlapping prediction set. Section 2 describes the pipeline; Section 3 reports our evaluation protocol, results, and an honest account of what we tried that did *not* work.

2 Methodology

2.1 Preprocessing

Raw GONG H-α observations are 2048×2048 single-channel intensity images. Each image is encoded into a 3-channel input so that the detector backbone can reuse standard RGB-pretrained weights while still operating on physically meaningful contrast bands: channel 1 is a percentile-clipped copy of the raw intensity; channel 2 divides the image by a large-kernel Gaussian estimate of the background (flattening limb darkening and the radial illumination gradient) followed by CLAHE; channel 3 is a black-hat morphological transform of channel 2, which isolates dark, thin, filament-scale detail. (A separate histogram-matching step is used only for the rejected pseudo-labeling experiment, not the main pipeline.)

2.2 Detection

Filament candidates are generated by **YOLO11m-seg** [5] (the medium-capacity variant), an anchor-free, NMS-based instance segmentation network, trained and run at 1280×1280 input resolution with a single foreground class. We trained one detector per cross-validation fold (Section 2.5) for up to 70 epochs (early-stopping patience 14) with batch size 4 and a fold-offset random seed (3407 + fold). Training was run with `deterministic=True` for reproducibility. Optimizer and augmentation were set explicitly rather than left at Ultralytics defaults: AdamW (lr_0=7×10^{-4}, lr_f=0.05, weight decay 5×10^{-4}, 3-epoch warmup); geometric augmentation (rotation $\pm180^\circ$, translate 0.04, scale 0.25, horizontal/vertical flip p=0.5 each); mosaic probability 0.20 (disabled for the final 10 epochs); hue/saturation jitter disabled (not meaningful for this derived contrast encoding) and value jitter 0.18; mask downsample ratio 2. At inference we use confidence threshold 0.26 and NMS IoU 0.55. The complete, machine-readable training call is in `scratch/`. We chose YOLO11-seg over a Hungarian-matching DETR-style detector (RF-DETR-Seg [7]) after a comparison on the same data split, preprocessing, and evaluator: even at its largest supported resolution, RF-DETR-Seg's raw PQ trailed YOLO11-seg by -0.057. This was not a matched-hyperparameter comparison (the two differ in resolution, epoch budget, and optimizer family), so it rules out RF-DETR-Seg as a drop-in replacement under its own defaults, not the architecture family in general.

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026
2026.

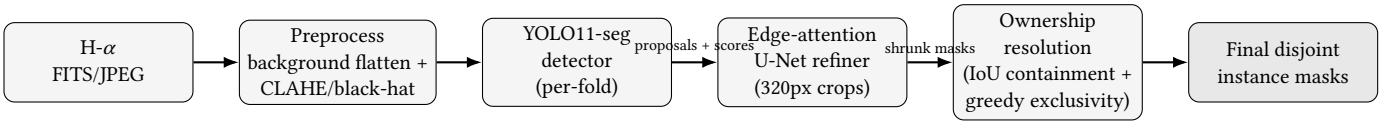


Figure 1: End-to-end inference pipeline. Training uses leak-free 3-fold cross-validation (Section 2.5); the submitted model uses a single detector fold plus a refiner fine-tuned on all folds.

2.3 Boundary Refinement

Raw detector masks are accurate in extent but imprecise at the pixel boundary, which is exactly where the official metric is least forgiving (Section 3.1). We refine each candidate independently: a square crop is taken around the candidate (1.65× its bounding box, minimum side 96px, 320 × 320px after resizing), passed through a U-Net refiner, and the resulting probability map is thresholded (0.45) and intersected with the original candidate mask (*shrink* refinement — the refiner may only remove pixels, not add them outside the detector's proposal).

Training data. Positive crops are generated one per ground-truth instance per annotator (so an image with 3 annotators contributes up to 3× as many crops as a single-annotator image), centered and scaled from that instance's bounding box using the same 1.65×-context rule as inference; we do not mine hard-negative (empty-target) crops in this recipe. Crops are augmented with random 90° rotations, horizontal/vertical flips, and a mild gamma jitter on the first two input channels.

Optimization. We minimize $0.35 \cdot \text{BCE} + 0.65 \cdot \text{soft-Dice} + 0.15 \cdot \text{boundary-}L_1$, where the boundary term compares a morphological-gradient (dilation−erosion, kernel 5) map of the prediction against the same map of the target, to bias the loss toward edge accuracy. We use AdamW (weight decay 10^{-4}), initial learning rate 2×10^{-4} with cosine annealing, automatic mixed precision, and batch size chosen per run as the largest of $\{8, 6, 4, 2\}$ that fits in GPU memory (8 in all runs reported here). Per-fold refiners train for 14 epochs with a fixed seed (3407); the checkpoint with the best validation crop-Dice (evaluated each epoch on the held-out fold, threshold 0.5) is kept. The submitted refiner is fine-tuned from the fold-0 checkpoint on all 707 images for 5 epochs at 0.35× the base learning rate (Section 2.5); with no held-out data at this stage, we keep the final-epoch weights rather than an epoch selected by a validation metric.

Our refiner, `MediSegEdgeAttUNet`, extends this base U-Net [8] with a single **edge-guided multi-head self-attention** (EG-MHSA) block at the bottleneck (8 heads), a mechanism *inspired by* EdgeAttNet [9]: a 3×3 convolution over the input crop produces a single-channel edge prior, which is projected and added to the query and key (but not the value) of a standard multi-head self-attention layer over the bottleneck features, before a residual connection and layer norm. EdgeAttNet's public code lacks a usable implementation of this module, so we could not validate ours against it, and our design deviates from the published one in ways that matter: EdgeAttNet stacks *two* EG-MHSA blocks with 4 heads each and reports that the edge prior lets it *drop* positional encodings and shrink to 22.7M parameters (versus 31.0M for its plain-U-Net baseline); our single-block, 8-head variant instead *adds* 4.2M parameters over our plain U-Net baseline (35.2M vs. 31.0M), matching the parameter count of what EdgeAttNet's own ablation calls "U-Net + MHSA, no positional encoding" almost exactly. We therefore call it edge-attention-inspired rather than a reproduction; Section 3 revisits what this parameter coincidence implies for our results.

2.4 Instance Resolution

Refined candidates can still overlap (two proposals covering the same physical filament, or a small proposal nested inside a larger one). We remove near-duplicates whose intersection covers ≥ 90% of the smaller candidate's area, then greedily assign remaining pixels to candidates in ascending order of area (small, precise candidates claim their pixels before large ones), discarding any candidate left with fewer than 20px. This ordering outperformed confidence-first and area-descending orderings in our ablations.

2.5 Cross-Validation and Final Model

The competition provides 707 training images, each annotated by one or more of several human experts (8,199 instance annotations in total, all annotators used as independent training targets). We group by unique observation filename (so leakage across the train and validation boundary of a fold is prevented by construction, independent of stratification) and partition the 707 filenames into three folds (236, 236, 235) with scikit-learn's stratified k-fold splitter (seed 3407), stratified on the quintile of each image's reference-annotator instance count so that sparse and dense images are balanced across folds. We train one detector and one refiner per fold, holding out that fold entirely. For the final submission we select the single detector fold with the best out-of-fold PQ, and fine-tune the refiner on *all* 707 images (warm-started from a held-out-verified checkpoint, 5 epochs at 0.35× the base learning rate) so the deployed refiner uses the full annotated set without invalidating our cross-validated model selection.

3 Evaluation

3.1 Evaluation Strategy

The competition's official metric is Panoptic Quality (PQ), following the definition of Kirillov et al. [6] but scored by the competition's own reference implementation [1], which we replicated locally against that notebook's test cases. Unlike the original PQ formulation's one-to-one Hungarian matching, the competition scorer counts *every* prediction-ground-truth pair with IoU above 0.5 as a true positive (i.e. both sides can be matched multiply), so

$$\text{PQ} = \frac{\sum_{(p,g)\in TP} \text{IoU}(p,g)}{|TP| + 0.5|FP| + 0.5|FN|}.$$

A prediction with IoU just under 0.5 scores zero *and* incurs a false-positive/false-negative penalty, so near-miss boundary quality dominates the score far more than raw detection recall. Matching is

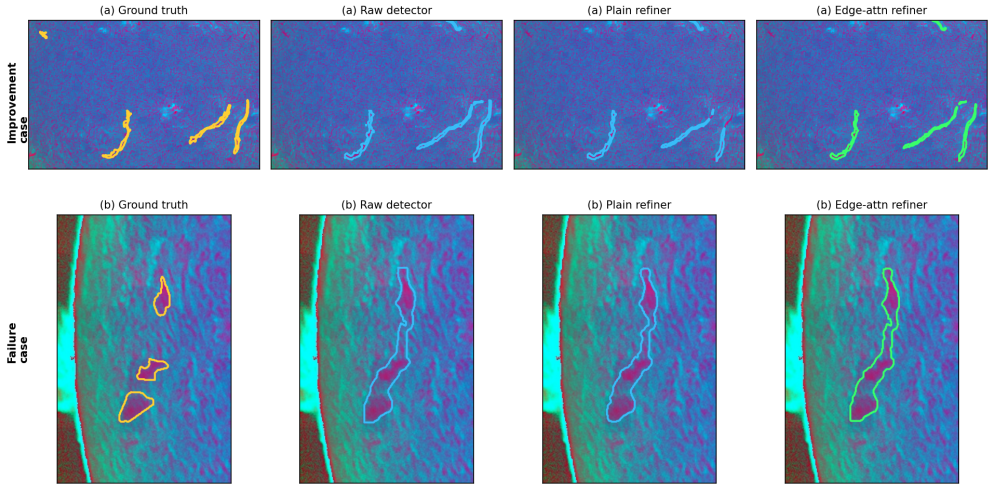


Figure 2: Ground truth vs. raw detector vs. plain refiner vs. edge-attention refiner, fold-0 OOF. (a) Improvement case: three filaments where the edge-attention refiner recovers substantially more true extent than the plain refiner (per-instance IoU $0.36 \rightarrow 0.59$**,** $0.43 \rightarrow 0.62$**,** $0.39 \rightarrow 0.56$**). (b) Failure case: three ground-truth instances where the detector yields one candidate straddling two (a merge error) and misses the third; all refiners inherit it — a detector-level failure refinement cannot fix. Both panels are from the original single-seed run; the ablation did not reproduce (a)'s comparison across seeds.**

done per (annotator, image) pair and micro-averaged, not averaged per-image.

Because human annotators themselves disagree, we measured a reference-annotation agreement baseline: treating one expert's annotations as the "prediction" and scoring them against every other expert who independently annotated the same image, under the identical official PQ rule, gives PQ= 0.3398 (multi-annotator images only, n=296 of 707; an earlier version trivially included self-matches, inflating the figure to 0.76 — corrected after review). We do not treat this as a hard ceiling on model performance (ours scores above it) — it is a benchmark for labeling-task reproducibility under the same matching rule.

All model-selection decisions use the leak-free 3-fold cross-validation of Section 2.5. To avoid a failure mode we caught twice during development — a change that improves one fold but does not replicate on the other two — we additionally compute an observation-cluster bootstrap [4] (2000 resamples, PQ as a ratio-of-sums rather than a mean-of-ratios) and only accept a change when both (a) its sign agrees across all folds and (b) its pooled 95% confidence interval excludes zero. Criterion (a) alone would be weak: under a symmetric null (no true effect), the probability that three independent folds all land on the same sign by chance is $2 \times 0.5^3 = 0.25$ two-sided (0.125 one-sided), so fold-sign-agreement by itself is only a screening heuristic, not a significance test; requiring (b) as well is what gives the combined criterion statistical weight. We report the pooled interval below for the one change that was accepted (edge-attention refiner, which satisfies both), while rejected changes are summarized in Section 3.3 without individual intervals. We also do not apply a multiple-comparisons correction across the roughly dozen distinct ideas evaluated against the same 707 OOF images over the course of this project, which inflates the effective false-positive rate above the nominal 5% for any single interval; we flag this explicitly rather than adjust for it post hoc, since the ideas were not pre-registered.

3.2 Results

Table 1 summarizes our out-of-fold (OOF) results under this protocol, and the corresponding public leaderboard scores where we submitted a matching variant. Note that our OOF PQ (0.4426, plain refiner over all 707 images) exceeds the 0.3398 reference-annotation agreement baseline of Section 3.1: this is not evidence the model is more *correct* than a human annotator, but it is consistent with the model, trained on pooled labels of several annotators, converging toward a consensus boundary that matches any single held-out annotator more often than two annotators match each other.

The edge-attention refiner's original per-fold deltas against the plain U-Net refiner were +0.0062 (fold 0), +0.0009 (fold 1), +0.0012 (fold 2); all three agreed in sign and the pooled bootstrap excluded zero (95% CI [0.00091, 0.00476]), so under our stated criterion we accepted and submitted it. We now believe that acceptance attributed the effect to the wrong mechanism, as the ablation below shows.

Matched-parameter ablation. To isolate whether the edge prior itself drives this effect, or whether added bottleneck attention capacity suffices, we ran two runs (seeds 3407, 8407) on all three folds for three refiner variants: a plain U-Net baseline (31.04M params), bottleneck MHSA with no edge prior (35.24M params), and the edge-attention refiner (35.24M params, differing from the no-edge-prior version by only ~2,100 parameters — these two *are* matched). Per-fold mean OOF PQ (base / plain attention / edge attention): fold 0 — 0.4414/0.4413/0.4405; fold 1 — 0.4405/0.4426/0.4417; fold 2 — 0.4434/0.4458/0.4464. By the same fold-sign lens: attention-minus-base is positive in two of three folds (1, 2), roughly zero in fold 0, for both attention variants. But edge-minus-plain-attention is at or below zero in two of three folds and sign-inconsistent in the

Configuration	OOF PQ	SQ / RQ	Public LB
Raw detector only	0.3937	0.652 / 0.604	—
+ refiner + resolution	0.4426	0.678 / 0.653	0.37
+ edge-attention refiner	0.4454[†]	—[†]	**0.39**
Ablations submitted for direct comparison:			
3-fold ensemble (standard refiner)	—	—	0.37
3-fold ensemble (edge-attention)	—	—	0.38
Matched-param. ablation, all 3 folds, 2 seeds each[‡]:			
Plain U-Net (31.04M params)	0.4418	0.678 / 0.651	—
+ plain MHSA, no edge prior (35.24M)	0.4432	0.681 / 0.651	—
+ edge-guided MHSA (35.24M)	0.4429	0.678 / 0.653	—

Table 1: PQ, SQ, RQ leak-free; first two blocks over all 707 images (raw detector: TP/FP/FN = 4765/2824/3434; +refiner+resolution: 5082/2286/3117; PQ=SQ×RQ). [†]The single-seed 0.4454 (baseline 0.4426 plus Δ = +0.0028, 95% CI [0.00091, 0.00476]) is the value originally accepted under our protocol and submitted at Public LB 0.39; the ablation below was run afterward to check it and did not reproduce it (Section 3.2). [‡]Seeds 3407, 8407; per-seed values in `output/ablation/`. Public LB is Kaggle's public score on a subset of our 180 held-out test predictions; the private leaderboard (undisclosed until the competition ends) is the basis for final ranking.

third — no fold shows the edge prior clearly beating plain attention. Full results are in `output/ablation/`.

The originally reported improvement, Public LB 0.39 vs. 0.37 plain, does not survive the follow-up check; this single submission pair may reflect leaderboard noise. Re-introducing the 3-fold ensemble scored 0.38 (Table 1). An expected-IoU scorer reported +0.0034 exploratory OOF gain (95% bootstrap CI [+0.00045, +0.00642]), but 0.37 publicly vs. production's 0.39. Its upstream model caches were not nested by outer fold, so this is not unbiased stacked-model OOF or adoption evidence, nor proof of the cause of the leaderboard reversal.

A χ^2 test over file-name metadata found no train/test shift in observation year or GONG site code to explain the ∼0.06 OOF-to-LB gap ($p = 0.42/0.94$), though 180 test images give limited power.

Submission evidence. The authors recorded 0.39 public scores for plain-MHSA fold-2 and edge-attention fold-2 checkpoint variants (mean best-match IoU 0.94/0.93 vs. production), motivating production plus plain-MHSA as architecture-diverse candidates; reported Kaggle T4 inference was ∼1.1 s/image. These historical runs predate a verified input–code–model–split–output manifest, and an alternate split copy differs on 489/707 observations; the weights dataset therefore also ships the consumed fold file `grouped_folds_train.csv` (SHA-256 `febb3f67...`, 236/236/235-file OOF-cache agreement) alongside the pinned registry copy (`24367120...`) — retraining must use the former. The current reproduction notebook reproduces both selected submissions byte-identically in a clean Kaggle environment (SHA-256 prefixes `84f1b89466f988d9` and `7b3de4e78a543cf2`). Other participants' provenance and public scores were not independently verified; score bands cannot establish misconduct.

3.3 Strengths and Weaknesses

Strengths. Five candidates were rejected under confirmation across independent folds, bootstrap intervals, or a matched public-LB comparison (weighted mask fusion, threshold retuning, refiner retraining, pseudo-labeling, 3-fold ensembling). The one change that passed our original protocol — edge-guided attention — did not survive a matched-parameter, multi-seed check across all three folds (Section 3.2): the ablation implicates attention capacity in general, not the edge prior. Our most useful finding is this refinement: fold-sign agreement plus a bootstrap CI could not attribute the effect to the right mechanism; only a matched-parameter, multi-fold follow-up could.

Weaknesses. Our original three-fold agreement conflated two mechanisms (attention capacity and the edge prior); the ablation is limited to two seeds per fold, and we lack direct evidence the edge prior is harmful — only that it adds no value beyond plain attention. Near-miss boundary errors (IoU 0.3–0.5) remain a major actionable error mode: fixing all 955 such OOF pairs would raise OOF PQ to ∼0.50; hard misses are more numerous (2,138 FN, 1,561 FP), and Figure 2(b) shows an instance-separation failure boundary refinement cannot address. The residual OOF-to-LB gap is unexplained beyond two ruled-out distribution-shift confounds (Section 3); an unverified public sparse-label model (∼77% retention) fits our counts to display precision. Detector scaling (larger backbone, multi-seed ensembling) was uniformly negative.

A later sweep targeted these error masses. Fold-2 error decomposition attributes ∼68% of false negatives to detector-blind cases (no proposal at any confidence); post-hoc scoring has a low ceiling (≈ +0.046). Twenty further mechanisms — EMA weight averaging, multi-channel refiner inputs, skeleton-based gap closure, dense-scene oversampling, a confidence sweep, a distance-transform loss, copy-paste augmentation, a SAM2-based promptable refiner, a proposal selector, detector-consensus and refiner flip-4 TTA, checkpoint ensembling, sub-threshold proposal rescue, an RF-DETR proposal-source union, temporal-neighbor augmentation, soft annotator-mean refiner targets (mean ΔPQ = −0.0034 over three folds, two seeds), union-label detector retraining on multi-annotator mask unions (fold-2 ΔPQ = −0.014 hard union, −0.004 agreement-weighted; the recovered detector-blind misses were outweighed by extra false positives), and self-supervised GONG Hα pretraining (SimCLR, 1,707 unlabeled frames, ±360 min train/test exclusion; verifier/refiner-encoder variants, fold-2 refiner PQ 0.4412 vs. scratch 0.4454, replicated at a second seed (−0.0003) and worse at 200 epochs (−0.022)) — all failed the frozen-operating-point gate (ΔPQ < +0.005). TTA, ensembling, and rescue carried signal but did not transfer (consensus +0.0007; flip-4 +0.0008; 3-ckpt +0.0021; rescue +0.0018); RF-DETR covered 317/612 detector-blind annotations at box IoU > 0.5 yet flooded the ownership stage (−0.028); temporal-neighbor augmentation tripled detector data yet cut recall (−0.009; stale labels, not signal). Co-temporal GONG magnetograms and polarity-inversion-line proximity were ignored (+0.000). Native 2×2-tile inference (1152 px) rescued 60/143 detector-blind truths to IoU ≥ 0.5 but

doubled false positives (ΔPQ = -0.033); a crop-level CNN verifier scored fold-2 AUC 0.64 vs. confidence 0.80; a public dossier (`too.foo`, LB 0.36) concurs. Three final mechanisms followed a public competitor diagnosis: classification-loss reweighting (cls = 1.5; fold deltas $-0.0037/+0.0012/+0.0038$ — fold-variance noise, replicated on a second platform), a semantic U-Net proposal source (`tu-convnext_tiny`, focal+Dice; union -0.0307: +61 blind truths vs. +524 false positives), and a 30-configuration strictness sweep on the cached maps (best -0.0076; rescue-scorer converging to baseline, -0.0009) — the residual miss pool is effectively blind to both detection and pixel classification at usable precision.

Two final mechanisms probed whether that residual mass is fixable at all. A Cellpose-style flow-field instance separator, trained on the same masks and fused as a further proposal source, reached the same frontier as the semantic head: it covered 163 of 953 held-out false negatives (17.1%, vs. the semantic source's 157, 16.5%) yet its union arm flooded the ownership stage (ΔPQ = -0.021 union, -0.076 proposal-only) — the detector, semantic, and flow-field proposal mechanisms each recovering the same ∼17% layer indicates that layer is bounded by the labels, not by model class. A direct measurement supports this: across the 296 multi-annotator images, pairs of experts agree at mean matched IoU 0.634 (median 0.630, n=2146 matched pairs) — below our model's measured SQ of 0.680 — while a typical annotator record recovers only 56.3% of the objects any other annotator marked. Roughly 44% of all instance annotations are single-annotator objects whose mean contrast (9.6 vs. 11.6) and visibility-flag rate (46% vs. 66%) mark them as judgment calls rather than invisible structures. Under per-(annotator, image) scoring, predicting such an object earns one true positive but ∼1.44 false positives against the other records on the same image, so even a perfect union-of-all-annotators predictor is bounded near PQ ∼0.49. A preregistered test of the remaining label-structure lever — a two-class detector labeling each instance *weak* (single-annotator) or *strong* (multi-annotator consensus), trained identically to production and swept over five weak-class score weights × four confidence floors — scored below production in all twenty cells (best ΔPQ = -0.041 at the primary operating point): the class head did separate the strata (strong-class recall 0.73 vs. weak 0.54) but sacrificed the majority weak class (76% of instances), and down-weighting it further was strictly worse. Together with the union-label variants this closes the label-structure line: per-record raw labels are optimal supervision for this metric, and the residual gap to a hypothetical ceiling is annotator disagreement, not model capacity. **Future work:** more seeds; further gains appear bounded by the labeling protocol itself.

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